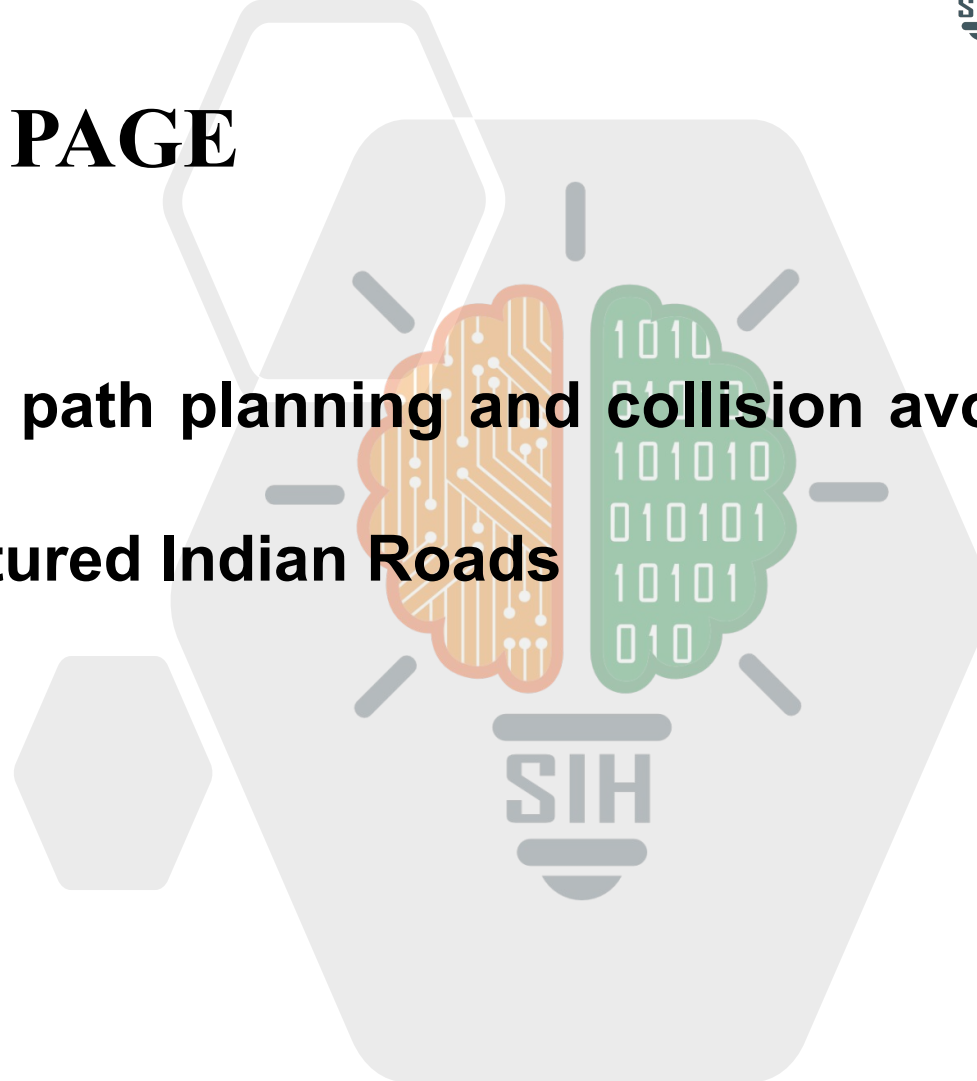
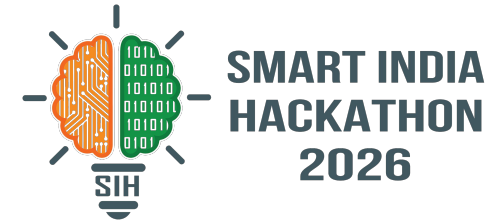


## TITLE PAGE

- Problem Statement ID – SIH26037
- Problem Statement Title- Adaptive path planning and collision avoidance in autonomous vehicle on unstructured Indian Roads
- Theme- Smart Vehicles
- PS Category- Software
- Team ID-
- Team Name - IQ Holders

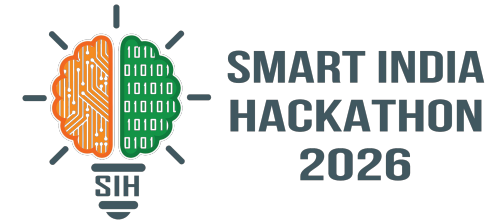


# IDEA TITLE



- **Problem Context:** The solution addresses the "Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads" challenge under the Smart Vehicles theme.
- **Core Concept:** A dynamic, closed-loop simulation pipeline that abandons strict lane-based assumptions in favor of probabilistic, real-time trajectory mapping.
- **Uniqueness:** The system specifically predicts the irregular, short-term motion of uniquely Indian road agents (e.g., auto-rickshaws, pushcarts, and pedestrians) and instantly recalculates safe paths when faced with sudden obstacles or missing lane markings.

# TECHNICAL APPROACH



- **Scenario Generation:** Use MathWorks RoadRunner to design the five critical test environments, including unmarked village roads, dense markets, and busy urban intersections without signals.
- **Perception Pipeline:** Integrate the Automated Driving Toolbox for sensor fusion (combining simulated camera, LiDAR, and radar data) and the Deep Learning Toolbox to detect diverse traffic using datasets like the Indian Driving Dataset (IDD).
- **Decision & Planning:** Implement the Navigation Toolbox to generate collision-free paths and Stateflow to design the state-machine logic required to handle abrupt events like sudden cattle crossings or informal highway merging.
- **Vehicle Dynamics:** Use the Vehicle Dynamics Blockset (or a Simulink bicycle model) to accurately simulate vehicle physics and validate metrics like replanning latency and path smoothness.

# FEASIBILITY AND VIABILITY



## Feasibility

### Software-Only Execution

The software-only category makes this highly feasible, as the entire perception-to-planning pipeline can be built and validated within the MATLAB/Simulink ecosystem.



## Challenges

### Replanning Latency

A major technical hurdle is keeping replanning latency low when predicting multiple erratic agents in dense traffic.



## Mitigation

### Model Optimization

This can be overcome by optimizing deep learning models for inference speed and using simplified bounding-box predictions for less critical peripheral agents.

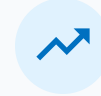
# IMPACT AND BENEFITS



## Social & Economic Benefit

### Safety-First ADAS Adaptation

Safely adapts autonomous driving technology to India's unique, unstructured road network, drastically reducing collision risks.

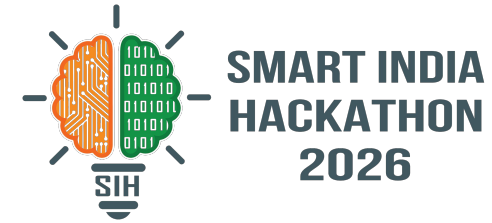


## Scalability

### Validated Testing Framework

Provides a validated, localized testing framework that automotive manufacturers can use to deploy Advanced Driver Assistance Systems (ADAS) tailored for developing nations.

# RESEARCH AND REFERENCES



***"IDD: A Dataset for Exploring Problems of Autonomous Navigation in Unconstrained Environments"*** (Varma et al., WACV): This is the primary academic citation required if you utilize the IDD dataset for training your perception models.

***"Explainable AI in Scene Understanding for Autonomous Vehicles in Unstructured Indian Roads"*** (MDPI): This paper proposes practical deep learning architectures for semantic segmentation on unstructured Indian roadways, specifically focusing on accurately segmenting drivable versus non-drivable areas in the absence of lane markings.

***"Autonomous Ground Robot in Unstructured Environments: How Far Have We Come?"*** (Journal of Field Robotics, 2026): A comprehensive, up-to-date survey covering over 250 recent papers on environmental perception, real-time path planning, and non-lane-based end-to-end driving.